FEAAM GmbH

Novel E-Motor Technologies

Drive Innovations

FEAAM GmbH
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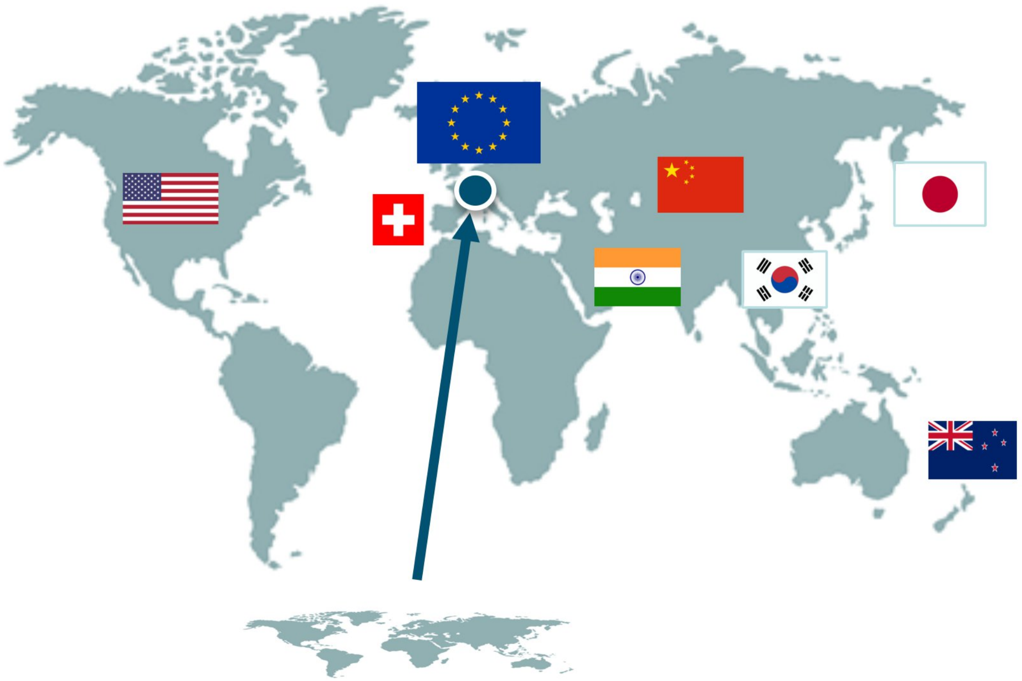
FEAAM: Deep-Tech Innovation in Electric Drives

The Challenge
- Strong dependency on rare-earth materials (cost volatility, supply risk)
- Limited efficiency gains in modern electric drives

What FEAAM Delivers
- Reduction or elimination of rare-earth magnets
- Increased torque density and efficiency through advanced motor design
- AI models trained on machine-specific electrical response

Why FEAAM
- Some 80 Patents worldwide
- Spin-off of Bundeswehr University Munich
- Technology excellence since foundation in 2006
- Team members rank among the Top 2% scientists worldwide[1)]
- Industrial partners around the world
- Global products with customized FEAAM designs inside

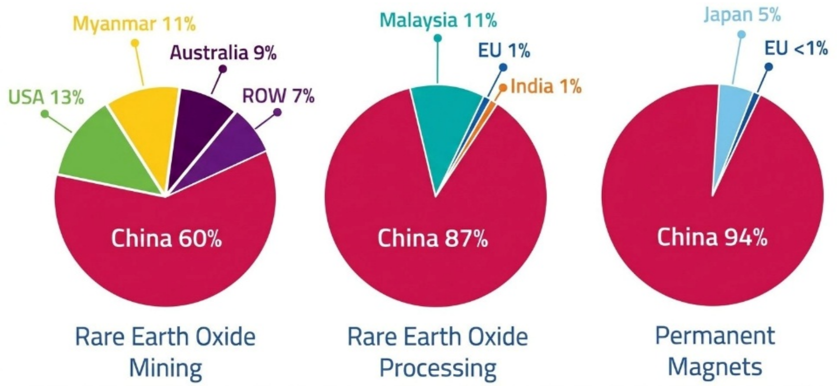


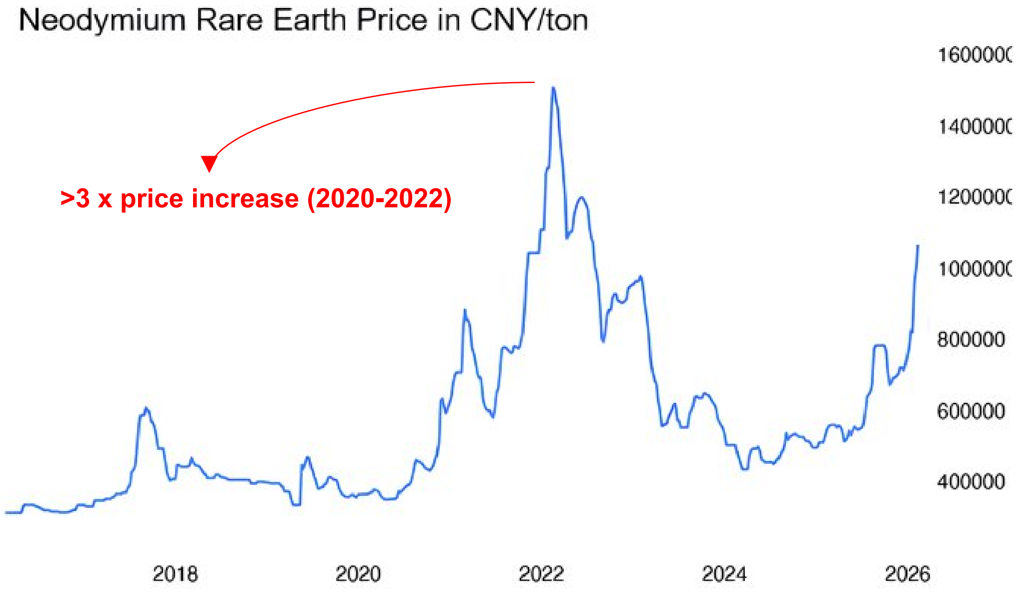
[1)] https://elsevier.digitalcommonsdata.com/

Rare-Earth Dependency: A Global Strategic Risk

🔗 Supply Chain Concentration

- 🗲60% of global rare-earth mining and 🗲 90% of processing concentrated in China
- >90% of permanent magnet production depends on this supply chain





🔗 Busi

- Material price volatility 🎭 margin exposure
- Supply dependency 🎭 production risk
- Electrification scaling 🎭 limited by material availability

Electrification Scale is Directly Limited by Rare-Earth Availability

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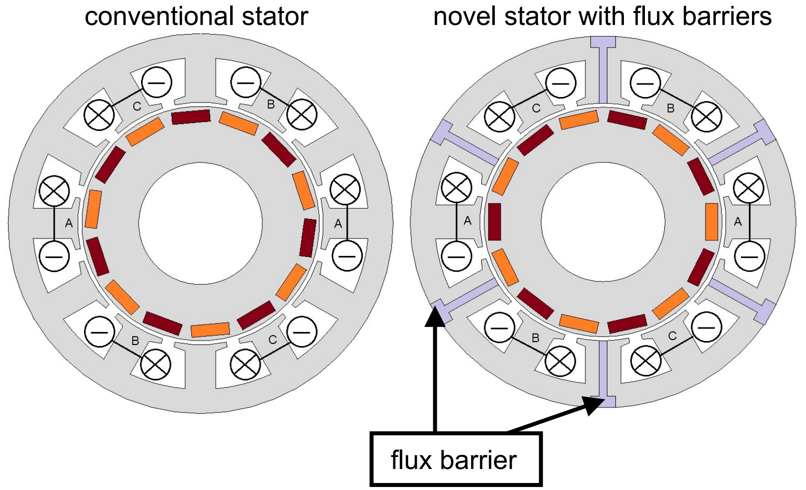
source: https://www.mining-technology.com/features/rare-earth-power-china-europe/?cf-view; https://tradingeconomics.com/commodity/neodymium
https://www.goldsilber.org/neodym-investment-eigenschaften-vorkommen-preis.php

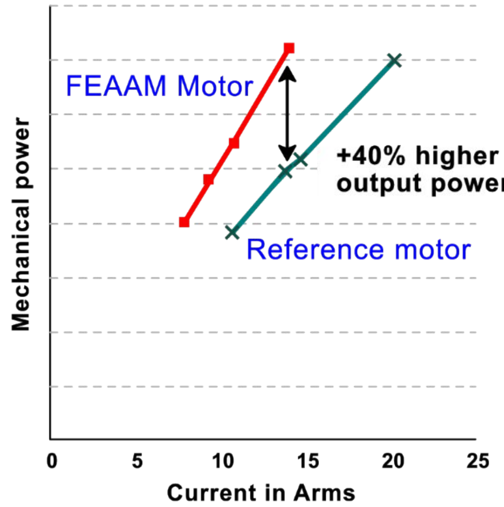
Solution: Reducing and Eliminating Rare-Earth Dependency

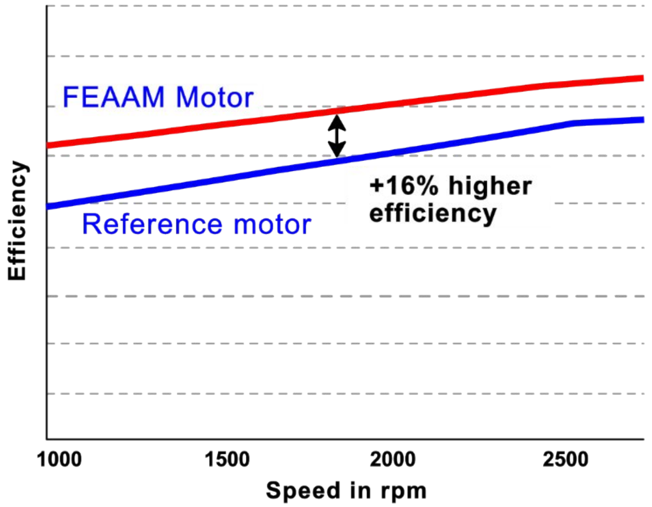
Flux Barrier Stator Technology:

- Reduced magnet material with same or higher performance
- High torque density in compact designs
- Improved efficiency and field weakening performance
- Proven technology in series production



Measurement results





FEAAM Technology Reduces Permanent Magnet Usage

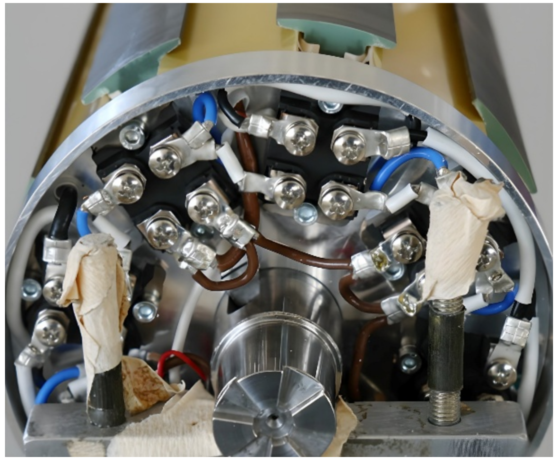
Fanatec® Direct Drive Pro Steering Wheel Motor

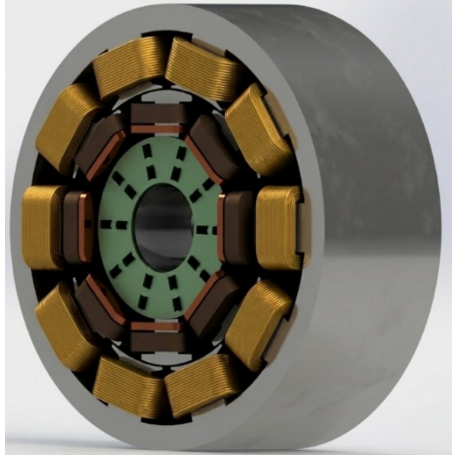
gtddpro.fanatec.com/gran-turismo-dd-pro-german/

Solution: Reducing and Eliminating Rare-Earth Dependency

Magnet - Free Motor Concept:

- High torque capability at zero speed
- No rare-earth magnets → complete supply independence
- Brushless design → no wear components (no slip-rings)
- Power density equivalent to motors with rare-earth magnets
- Suitable for high-performance traction systems

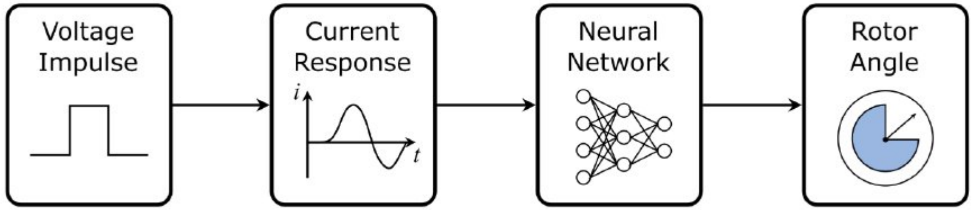




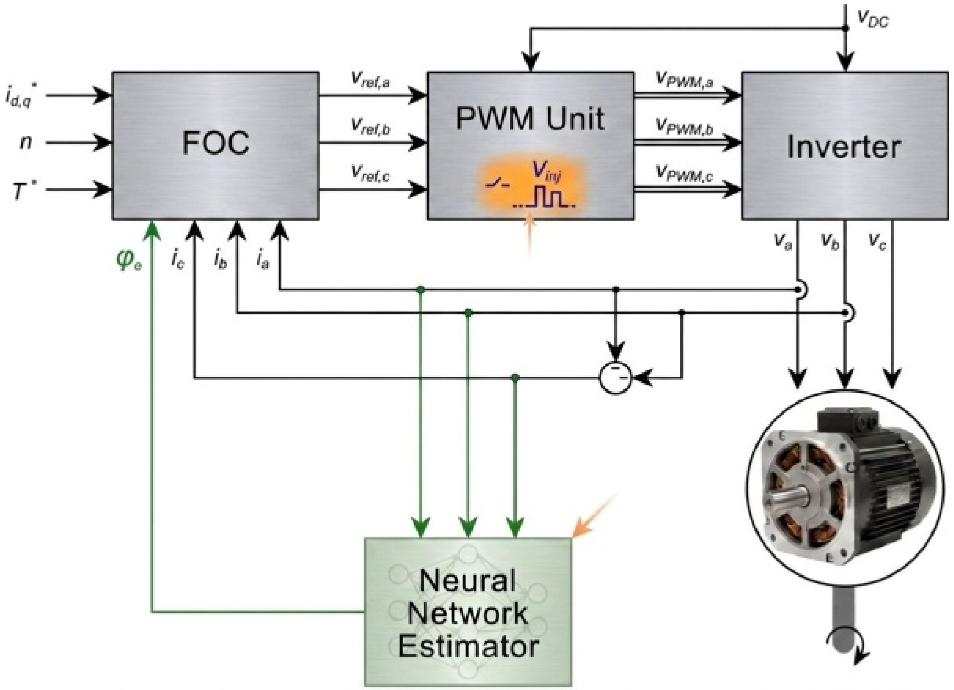
Full Decoupling from Supply Chain Risk Resulting from 0% Rare-Earth Materials

AI-Based Sensorless Rotor Position Estimation

- Core Concept:
  - Voltage pulse injection at standstill and low speeds
  - AI reconstructs rotor angle from current response
  - Works where conventional methods fail (standstill and low speed)



- Key Advantages
  - No mechanical position sensors 🎭 lower cost and complexity
  - Robust operation (no sensor failure)
  - Real-time capable on embedded systems



Accurate Rotor Position Estimation Without Sensors, even at Zero Speed

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